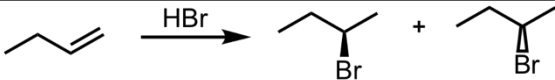
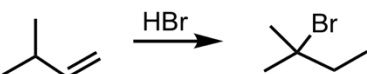
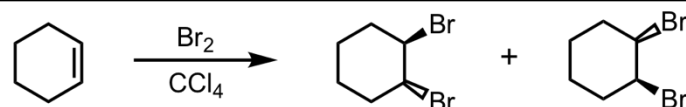
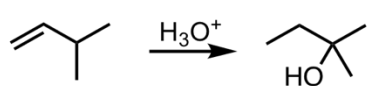
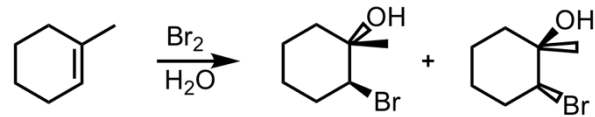
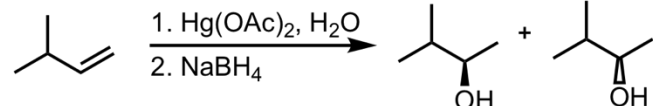
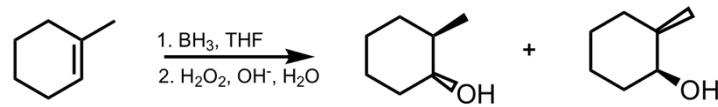
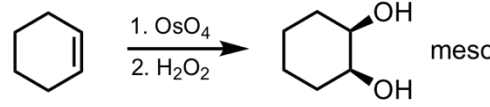
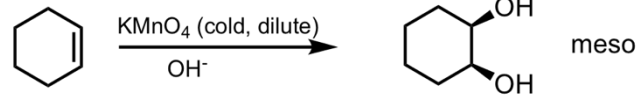
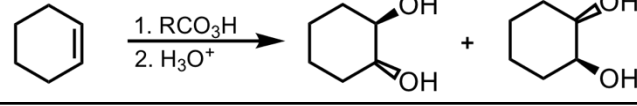
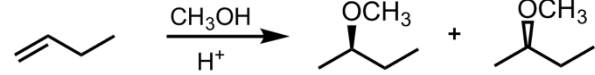
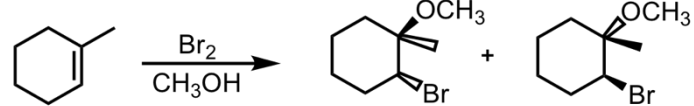


Bootcamp.com Organic Chemistry Reaction Summary Sheet

Alkene Reactions

Hydrohalogenation	
Hydrohalogenation (with Rearrangement)	
Halogenation	
Hydrobromination with Peroxide	
Hydration	 *any acid catalyst (H ₂ SO ₄ , H ₃ PO ₄ , etc.) and water
Hydration (with Rearrangement)	
Bromination in H ₂ O	
Oxymercuration-Demercuration	
Hydroboration-Oxidation	
Syn-Dihydroxylation	
Syn-Dihydroxylation	
Anti-Dihydroxylation	
Addition of an Alcohol	
Bromination in Alcohol	

Alkoxymercuration-Demercuration	
Epoxidation	
Catalytic Hydrogenation	meso *Pt can also be used*
Ozonolysis (Reducing Conditions)	
Ozonolysis (Oxidizing Conditions)/Oxidative Cleavage	

Alkyne Reactions

Catalytic Hydrogenation (Catalytic Reduction)	
Reduction to Cis-Alkene	
Reduction to Trans-Alkene	
Hydrohalogenation with HBr (Terminal Alkyne)	
Hydrohalogenation with HBr (Internal Alkyne)	
Halogenation with Br ₂	
Hydration of an Internal Alkyne	
Hydration of a Terminal Alkyne (Markovnikov)	
Hydration of a Terminal Alkyne (Anti-Markovnikov)	

S_N2 Addition of an Acetylide Ion to an Alkyl Halide	$\text{HC}\equiv\text{CH} \xrightarrow{\text{NaNH}_2} \text{HC}\equiv\text{C}^- \xrightarrow{\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}} \text{HC}\equiv\text{CCH}_2\text{CH}_2\text{CH}_3$
S_N2 Addition of an Acetylide Ion to a Ketone	$\text{HC}\equiv\text{CH} \xrightarrow{\text{NaNH}_2} \text{HC}\equiv\text{C}^- \xrightarrow[2. \text{H}_2\text{O}]{1. \text{CH}_3\text{COCH}_3} \text{HC}\equiv\text{CCH}(\text{CH}_3)_2\text{OH}$
S_N2 Addition of an Acetylide Ion to an Epoxide	$\text{HC}\equiv\text{CH} \xrightarrow{\text{NaNH}_2} \text{HC}\equiv\text{C}^- \xrightarrow[2. \text{H}_2\text{O}]{1. \text{propylene oxide}} \text{HC}\equiv\text{CCH}_2\text{CH}(\text{CH}_3)\text{OH}$
Ozonolysis/Oxidative Cleavage on an Internal Alkyne	$\text{CH}_3\text{CH}_2\text{C}\equiv\text{CCH}_3 \xrightarrow[2. \text{H}_2\text{O}_2]{1. \text{O}_3} \text{CH}_3\text{CH}_2\text{COOH} + \text{CH}_3\text{COOH}$ $\text{or} \quad \xrightarrow[2. \text{H}_3\text{O}^+]{1. \text{KMnO}_4/\text{heat}}$
Ozonolysis/Oxidative Cleavage on a Terminal Alkyne	$\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH} \xrightarrow[2. \text{H}_2\text{O}_2]{1. \text{O}_3} \text{CH}_3\text{CH}_2\text{COOH} + \text{CO}_2$ $\text{or} \quad \xrightarrow[2. \text{H}_3\text{O}^+]{1. \text{KMnO}_4/\text{heat}}$ Terminal carbon becomes CO_2
Alkyne Formation from Double Elimination of a Vicinal Dihalide	$\text{CH}_3\text{CH}(\text{Br})\text{C}(\text{Br})(\text{CH}_3)\text{CH}_3 \xrightarrow{\text{NaNH}_2} \text{CH}_3\text{C}(\text{Br})=\text{CHCH}_3 \xrightarrow{\text{NaNH}_2} \text{CH}_3\text{C}\equiv\text{CCH}_3$

Free Radical Halogenation Reactions

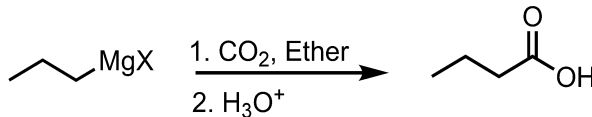
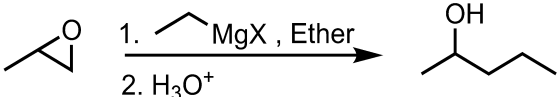
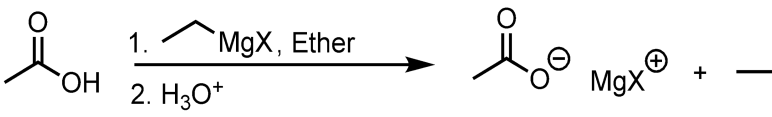
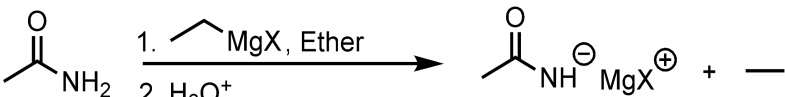
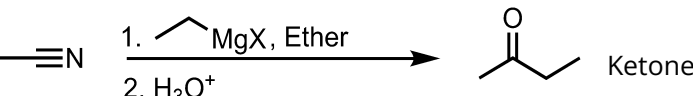
Free Radical Halogenation using Bromine (more selective)	$\text{1-methylcyclopentane} \xrightarrow[hv \text{ or } \Delta]{\text{Br}_2} \text{1-bromo-1-methylcyclopentane}$
Free Radical Halogenation using Chlorine (less selective)	$\text{1-methylcyclopentane} \xrightarrow[hv \text{ or } \Delta]{\text{Cl}_2} \text{1-chloro-1-methylcyclopentane} + \text{1-chloro-2-methylcyclopentane} + \text{2-chloro-1-methylcyclopentane} + \text{3-chloro-1-methylcyclopentane}$
Allylic/Benzylic Bromination	$\text{toluene} \xrightarrow[hv \text{ or } \Delta \text{ or ROOR}]{\text{NBS}} \text{benzyl bromide}$ $\text{1-butene} \xrightarrow[hv \text{ or } \Delta \text{ or ROOR}]{\text{NBS}} \text{3-bromobut-1-ene} + \text{1-bromo-2-butene}$

Diels-Alder Reactions

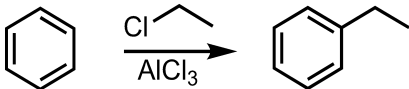
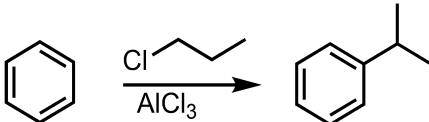
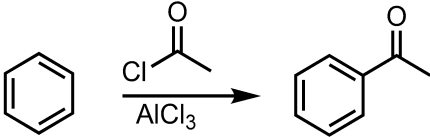
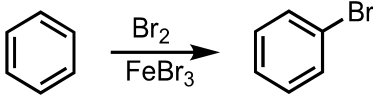
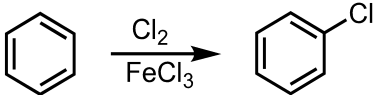
Diene Addition to a Dienophile (Alkene)	
Diene Addition to a Dienophile (Alkyne)	
Diene Addition to a <i>cis</i> Dienophile	
Diene Addition to a <i>trans</i> Dienophile	
Diene Addition to a substituted Dienophile	

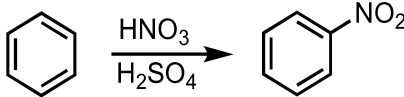
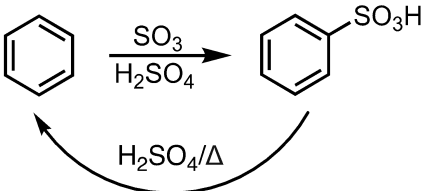
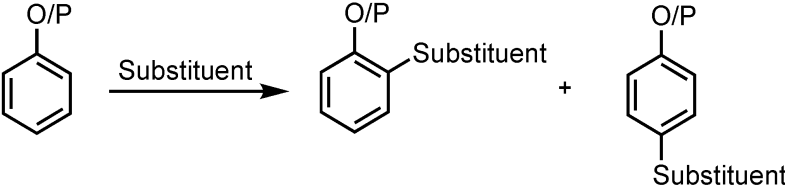
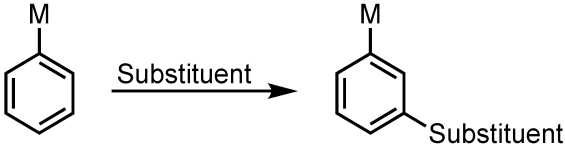
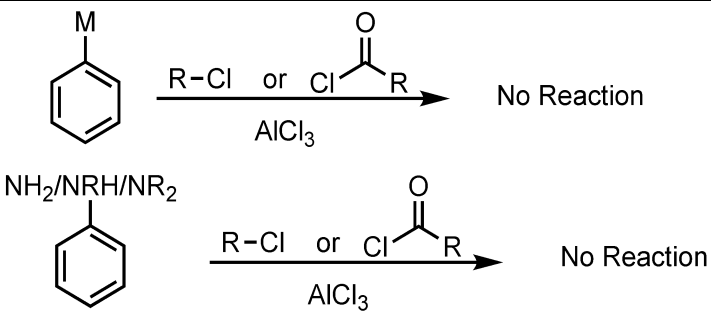
Grignard Reactions

Addition of a Grignard Reagent to an Aldehyde	
Addition of a Grignard Reagent to a Ketone	
Addition of a Grignard Reagent to an Ester	
Addition of a Grignard Reagent to an Acyl Chloride	

Addition of a Grignard Reagent to CO ₂	 <chem>CC[Mg]X.CC(=O)O</chem> Carboxylic Acid
Addition of a Grignard Reagent to an Epoxide (adds to the less subs. side)	 <chem>CC[Mg]X.CCO</chem>
Addition of a Grignard Reagent to a Carboxylic Acid	 Carboxylate
Addition of a Grignard Reagent to an Amide	 Deprotonated Amide
Addition of a Grignard Reagent to a Nitrile	 Ketone

Electrophilic Aromatic Substitution (EAS) Reactions

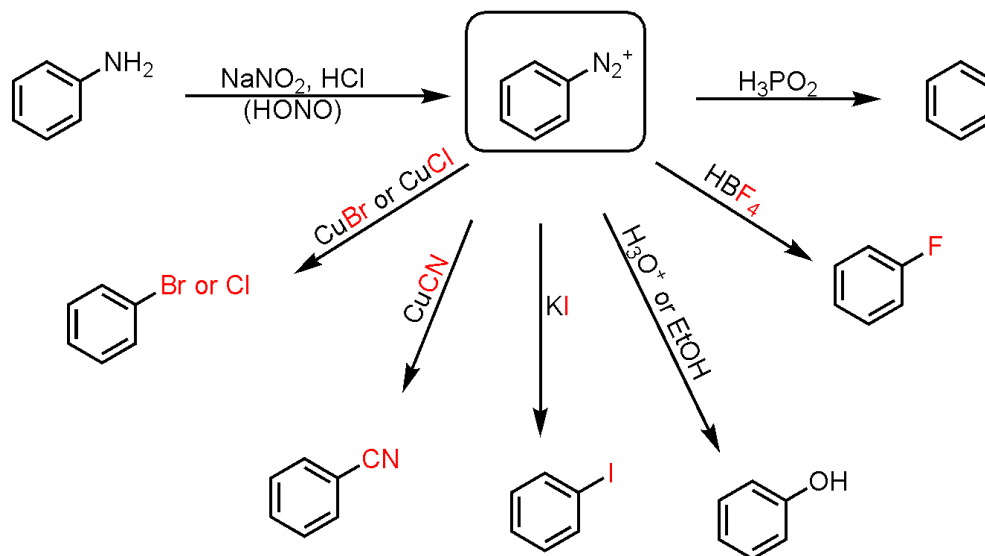
Friedel-Crafts Alkylation (Rearrangement Possible)	 
Friedel-Crafts Acylation (No Rearrangement Possible)	
Bromination	
Chlorination	

Nitration	
Sulfonation	
Formylation	
EAS with an ortho/para-directing group on Benzene	
EAS with a meta-directing group on Benzene	
Friedel-Crafts Alkylation/Acylation with a meta-directing group or an amine on Benzene	

Benzene Side-Chain Reactions

Side-Chain Oxidation of Benzene to form Benzoic Acid	<chem>c1ccccc1C</chem> or <chem>c1ccccc1CR</chem> or <chem>c1ccccc1C(R)R</chem> $\xrightarrow[2. \text{H}_3\text{O}^+, \text{Heat}]{1. \text{KMnO}_4, ^-\text{OH}}$ <chem>c1ccccc1C(=O)O</chem> or $\xrightarrow{\text{Na}_2\text{Cr}_2\text{O}_7, \text{H}_2\text{SO}_4}$ <chem>c1ccccc1C(C)(C)C</chem> $\xrightarrow[2. \text{H}_3\text{O}^+, \text{Heat}]{1. \text{KMnO}_4, ^-\text{OH}}$ or $\xrightarrow{\text{Na}_2\text{Cr}_2\text{O}_7, \text{H}_2\text{SO}_4}$ No Reaction (Requires free Hydrogen at Benzylic position)
Wolff-Kishner Reduction	<chem>c1ccccc1C(=O)C</chem> $\xrightarrow{\text{H}_2\text{NNH}_2 \text{ or } \text{N}_2\text{H}_4, ^-\text{OH}, \text{Heat}}$ <chem>c1ccccc1CC</chem>
Clemmensen Reduction	<chem>c1ccccc1C(=O)C</chem> $\xrightarrow{\text{Zn(Hg), HCl, Heat}}$ <chem>c1ccccc1CC</chem> *can also use H₂/Pd, C <chem>c1ccccc1[N+](=O)[O-]</chem> $\xrightarrow{\text{Zn(Hg), HCl, Heat}}$ <chem>Nc1ccccc1</chem> *can also use H₂/Pd, C or Sn/HCl
Acetylation of Aniline using Acetic Anhydride	<chem>Nc1ccccc1</chem> (Aniline) $\xrightarrow{\text{pyridine}}$ <chem>CC(=O)Nc1ccccc1</chem> (Acetanilide)

Diazonium Salt Reactions



Hydride Reduction Reactions

Reduction of an Aldehyde to a 1° Alcohol	$\text{CH}_3\text{CHO} \xrightarrow[2. \text{H}_3\text{O}^+]{1. \text{NaBH}_4} \text{CH}_3\text{CH}_2\text{OH}$ $\text{CH}_3\text{CHO} \xrightarrow[2. \text{H}_3\text{O}^+]{1. \text{LiAlH}_4} \text{CH}_3\text{CH}_2\text{OH}$
Reduction of a Ketone to a 2° Alcohol	$\text{CH}_3\text{COCH}_3 \xrightarrow[2. \text{H}_3\text{O}^+]{1. \text{NaBH}_4} \text{CH}_3\text{CH(OH)CH}_3$ $\text{CH}_3\text{COCH}_3 \xrightarrow[2. \text{H}_3\text{O}^+]{1. \text{LiAlH}_4} \text{CH}_3\text{CH(OH)CH}_3$
Reduction of a Carboxylic Acid to a 1° Alcohol	$\text{CH}_3\text{COOH} \xrightarrow[2. \text{H}_3\text{O}^+]{1. \text{LiAlH}_4} \text{CH}_3\text{CH}_2\text{OH}$
Reduction of an Ester to a 1° Alcohol	$\text{CH}_3\text{COOCH}_3 \xrightarrow[2. \text{H}_3\text{O}^+]{1. \text{LiAlH}_4} \text{CH}_3\text{CH}_2\text{OH} + \text{CH}_3\text{OH}$
Reduction of an Ester to an Aldehyde	$\text{CH}_3\text{COOCH}_3 \xrightarrow[2. \text{H}_2\text{O}]{1. \text{DIBAL-H, } -78^\circ\text{C}} \text{CH}_3\text{CHO}$

Reduction of an Acyl Chloride to a 1° Alcohol	$\text{CH}_3\text{COCl} \xrightarrow[2. \text{H}_3\text{O}^+]{1. \text{LiAlH}_4} \text{CH}_3\text{CH}_2\text{OH}$ $\text{CH}_3\text{COCl} \xrightarrow[2. \text{H}_3\text{O}^+]{1. \text{NaBH}_4, \text{EtOH}} \text{CH}_3\text{CH}_2\text{OH}$
Reduction of an Acyl Chloride to an Aldehyde	$\text{CH}_3\text{COCl} \xrightarrow{\text{LiAlH}[\text{OC}(\text{CH}_3)_3]_3} \text{CH}_3\text{CHO}$
Reduction of an Amide to an Amine	$\text{CH}_3\text{CH}_2\text{CONH}_2 \xrightarrow[2. \text{H}_3\text{O}^+]{1. \text{LiAlH}_4} \text{CH}_3\text{CH}_2\text{NH}_2$
Hofmann Rearrangement	$\text{CH}_3\text{CH}_2\text{CONH}_2 \xrightarrow[2. \text{NaOH}]{1. \text{Br}_2} \text{CH}_3\text{CH}_2\text{NH}_2$
Reduction of a Nitrile to an Amine	$\text{CH}_3\text{C}\equiv\text{N} \xrightarrow[2. \text{H}_3\text{O}^+]{1. \text{LiAlH}_4} \text{CH}_3\text{CH}_2\text{NH}_2$

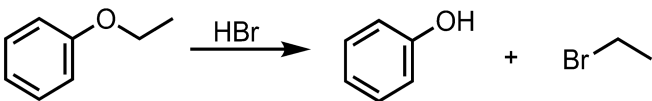
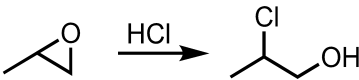
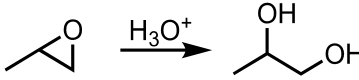
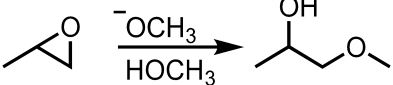
Alcohol Reactions

Conversion of a 2°/3° Alcohol to an alkyl halide via S _N 1	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_3 \xrightarrow{\text{HX}} \text{CH}_3\text{CH}(\text{X})\text{CH}_3$ $(\text{CH}_3)_3\text{COH} \xrightarrow{\text{HX}} (\text{CH}_3)_3\text{CX}$
Conversion of a 1°/2° Alcohol to an alkyl bromide via S _N 2	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \xrightarrow{\text{PBr}_3} \text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ $(\text{CH}_3)_2\text{CHCH}_2\text{OH} \xrightarrow{\text{PBr}_3} (\text{CH}_3)_2\text{CHCH}_2\text{Br}$
Conversion of a 1°/2° Alcohol to an alkyl chloride via S _N 2	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \xrightarrow[\text{Pyridine}]{\text{SOCl}_2} \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ $(\text{CH}_3)_2\text{CHCH}_2\text{OH} \xrightarrow[\text{Pyridine}]{\text{SOCl}_2} (\text{CH}_3)_2\text{CHCH}_2\text{Cl}$
Conversion of an Alcohol to a Tosylate Ester (OTs)	$(\text{CH}_3)_2\text{CHCH}_2\text{OH} \xrightarrow{\text{TsCl}} (\text{CH}_3)_2\text{CHCH}_2\text{OTs}$ Retention of Stereochemistry
Acid-catalyzed Dehydration of an Alcohol	$(\text{CH}_3)_2\text{CHCH}_2\text{OH} \xrightarrow{\text{H}_3\text{O}^+} (\text{CH}_3)_2\text{CHCH}=\text{CH}_2$ Zaitsev's Rule

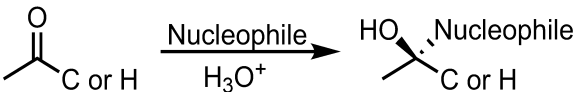
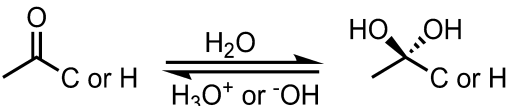
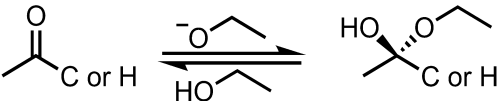
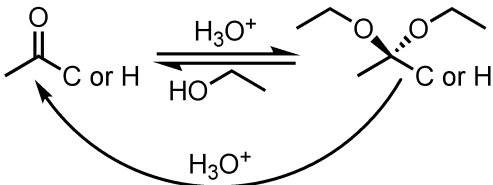
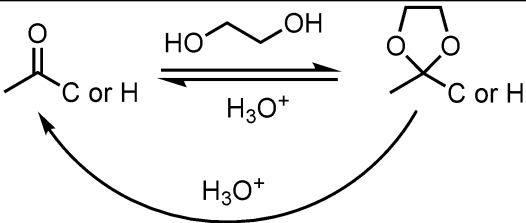
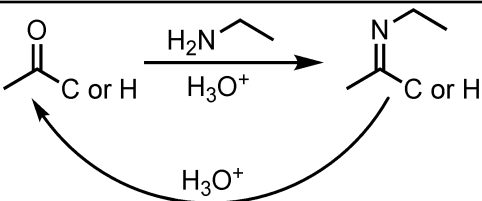
Chromic Acid Oxidation of a 1° Alcohol to a Carboxylic Acid	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \xrightarrow[\text{H}_2\text{SO}_4]{\text{Na}_2\text{Cr}_2\text{O}_7 \text{ or } \text{CrO}_3} \text{CH}_3\text{CH}_2\text{COOH}$
Chromic Acid Oxidation of a 2° Alcohol to a Ketone	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3 \xrightarrow[\text{H}_2\text{SO}_4]{\text{Na}_2\text{Cr}_2\text{O}_7 \text{ or } \text{CrO}_3} \text{CH}_3\text{CH}_2\text{COCH}_3$
Chromic Acid Oxidation of an Aldehyde to a Carboxylic Acid	$\text{CH}_3\text{CH}_2\text{CHO} \xrightarrow[\text{H}_2\text{SO}_4]{\text{Na}_2\text{Cr}_2\text{O}_7 \text{ or } \text{CrO}_3} \text{CH}_3\text{CH}_2\text{COOH}$
PCC or DMP Oxidation of a 1° Alcohol to an Aldehyde	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \xrightarrow{\text{PCC or DMP}} \text{CH}_3\text{CH}_2\text{CHO}$
PCC or DMP Oxidation of a 2° Alcohol to a Ketone	$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3 \xrightarrow{\text{PCC or DMP}} \text{CH}_3\text{CH}_2\text{COCH}_3$
Oxidative Cleavage of a 1,2 Diol	$\begin{array}{c} \text{HO}-\text{CH}(\text{CH}_3)-\text{CH}(\text{CH}_3)-\text{OH} \xrightarrow{\text{HIO}_4} \text{CH}_3\text{CHO} + \text{CH}_3\text{CHO} \\ \text{HO}-\text{CH}(\text{CH}_3)-\text{CH}(\text{CH}_3)-\text{OH} \xrightarrow{\text{HIO}_4} \text{CH}_3\text{COCH}_3 + \text{CH}_3\text{CHO} \end{array}$
Swern Oxidation	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \xrightarrow[2. \text{Et}_3\text{N}]{1. \text{DMSO}, (\text{COCl})_2 [\text{oxalyl chloride}]} \text{CH}_3\text{CH}_2\text{CHO}$

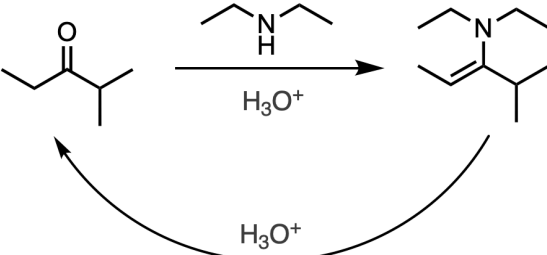
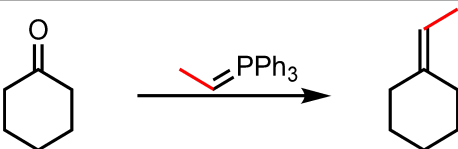
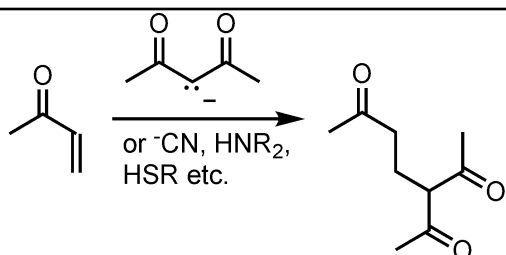
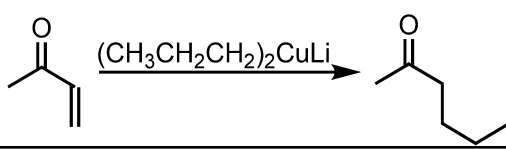
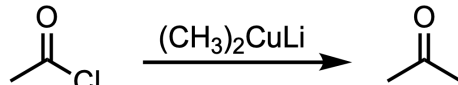
Ether and Epoxide Reactions

Williamson Ether Synthesis via S _N 2	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \xrightarrow{\text{NaH, Na, or K}} \text{CH}_3\text{CH}_2\text{CH}_2\text{O}^- \xrightarrow{\text{CH}_3\text{CH}_2\text{Cl}} \text{CH}_3\text{CH}_2\text{CH}_2\text{OCH}_2\text{CH}_3$
Acid-catalyzed Cleavage of Ethers when one side is 2°/3° (Nucleophile attacks more substituted side via S _N 1)	$\begin{array}{c} \text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3 \xrightarrow{\text{HBr}} \text{CH}_3\text{CH}_2\text{Br} + \text{CH}_3\text{CH}_2\text{OH} \\ \text{CH}_3\text{C}(\text{CH}_3)_2\text{OCH}_2\text{CH}_3 \xrightarrow{\text{HBr}} \text{CH}_3\text{C}(\text{CH}_3)_2\text{Br} + \text{CH}_3\text{CH}_2\text{OH} \end{array}$

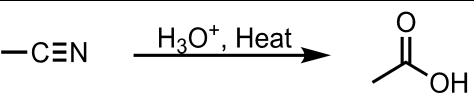
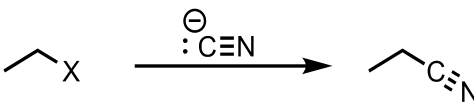
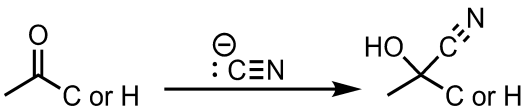
Acid-catalyzed Cleavage of Ethers when neither side is 2°/3° (Nucleophile attacks less substituted side via S _N 2)	
Acid-catalyzed Ring Opening of Epoxides (Nucleophile attacks more substituted side)	 
Base-catalyzed Ring Opening of Epoxides (Nucleophile attacks less substituted side)	

Aldehyde and Ketone Reactions

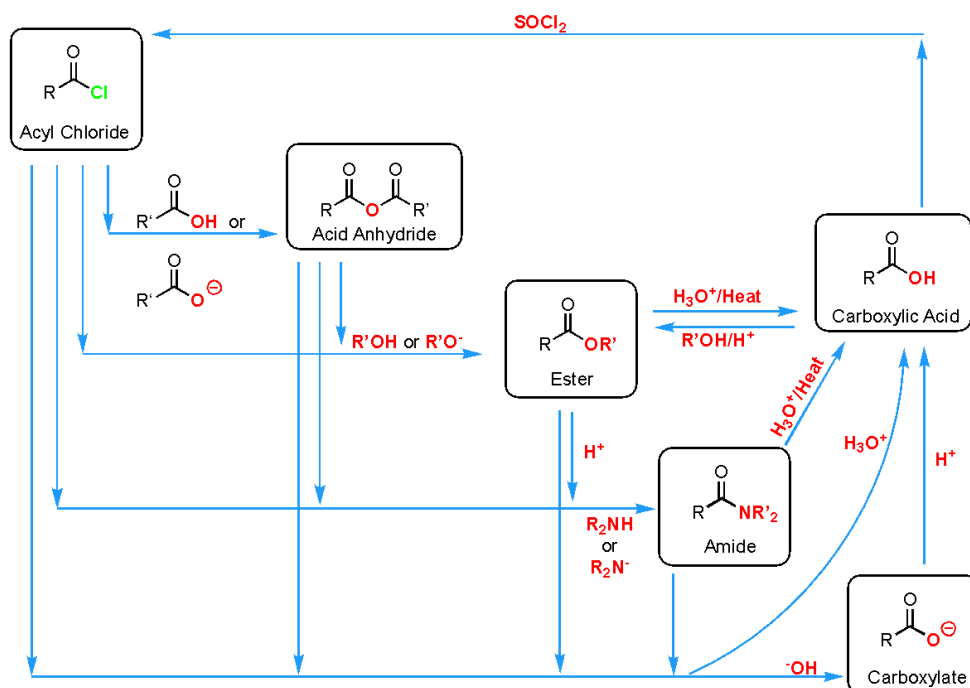
Nucleophilic Addition to an Aldehyde or Ketone	
Addition of water to an Aldehyde or Ketone forming a Hydrate	
Base-catalyzed addition of an Alcohol to an Aldehyde or Ketone forming a Hemi-acetal/Hemi-ketal	
Acid-catalyzed addition of an Alcohol to an Aldehyde or Ketone forming a Acetal/Ketal (Protecting Group, reversed by H ₃ O ⁺)	
Acid-catalyzed addition of Ethylene Glycol to an Aldehyde or Ketone forming a Acetal/Ketal (Protecting Group, reversed by H ₃ O ⁺)	
Addition of a 1° Amine to an Aldehyde or Ketone forming an Imine (Reversed by H ₃ O ⁺)	

Addition of a 2° Amine to an Aldehyde or Ketone forming an Enamine (Reversed by H_3O^+)	 Double bond forms on less substituted end for ketones
Addition of a Wittig Reagent to an Aldehyde/Ketone	
Michael Addition to an α, β Unsaturated Ketone	 or ^-CN, HNR_2, HSR etc.
Michael Addition to an α, β Unsaturated Ketone with a Gilman Reagent (Organocuprates)	
Formation of a Ketone via Nucleophilic Acyl Substitution with a Gilman Reagent (Organocuprate)	

Nitrile Reactions

Acid-catalyzed Hydrolysis of a Nitrile	
$\text{S}_\text{N}2$ formation of Nitriles using Cyanide and Alkyl Halides	
Cyanohydrin Formation using Aldehydes/Ketones and Cyanide	

Carboxylic Acid Derivative Reactions



Alpha Addition/Substitution Reactions

Self Aldol Condensation and Enone Formation	$2 \text{ CH}_3\text{CHO} \xrightarrow{^-\text{OH}, \text{H}_2\text{O}} \text{CH}_3\text{CH(OH)CH}_2\text{CHO} \xrightarrow[\Delta]{\text{H}_3\text{O}^+, \text{NaOH}} \text{CH}_3\text{CH=CHCHO}$ $2 \text{ CH}_3\text{COCH}_3 \xrightarrow{^-\text{OH}, \text{H}_2\text{O}} \text{CH}_3\text{C(OH)(CH}_3\text{)CH}_2\text{COCH}_3 \xrightarrow[\Delta]{\text{H}_3\text{O}^+, \text{NaOH}} \text{CH}_3\text{C(CH}_3\text{)=CHCOCH}_3$
Mixed Aldol Condensation and Enone Formation	$\text{CH}_3\text{COCH}_3 + \text{C}_6\text{H}_5\text{CHO} \xrightarrow{^-\text{OH}, \text{H}_2\text{O}} \text{CH}_3\text{C(OH)(CH}_3\text{)CH}_2\text{C}_6\text{H}_5 \xrightarrow[\Delta]{\text{H}_3\text{O}^+, \text{NaOH}} \text{CH}_3\text{C(CH}_3\text{)=CHC}_6\text{H}_5$ $2 \text{ CH}_3\text{CH}_2\text{COCH}_3 \xrightarrow{^-\text{OH}, \text{H}_2\text{O}} \text{CH}_3\text{CH}_2\text{C(OH)(CH}_3\text{)CH}_2\text{COCH}_3 \xrightarrow[\Delta]{\text{H}_3\text{O}^+, \text{NaOH}} \text{CH}_3\text{CH}_2\text{C(CH}_3\text{)=CHCOCH}_3$
Self Claisen Condensation	$2 \text{ CH}_3\text{COOCH}_2\text{CH}_3 \xrightarrow[2. \text{H}_3\text{O}^+]{1. ^-\text{OCH}_2\text{CH}_3} \text{CH}_3\text{COCH}_2\text{COOCH}_2\text{CH}_3$
Mixed Claisen Condensation	$\text{CH}_3\text{COCH}_3 + \text{CH}_3\text{COOCH}_2\text{CH}_3 \xrightarrow[2. \text{H}_3\text{O}^+]{1. ^-\text{OCH}_2\text{CH}_3} \text{CH}_3\text{COCH}_2\text{COCH}_3$

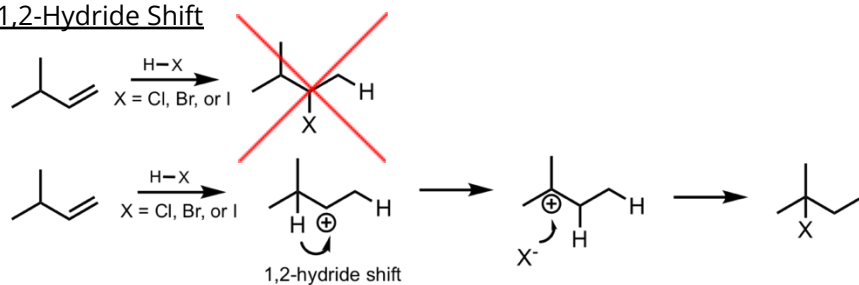
Dieckmann Cyclization (Intramolecular Claisen Condensation)	
Acetoacetic Ester Synthesis	
Malonic Ester Synthesis	
Alpha Halogenation In Basic Conditions	
Alpha Halogenation in Acidic Conditions	
Haloform Reaction	*A methyl group is required for this reaction

Reaction Details

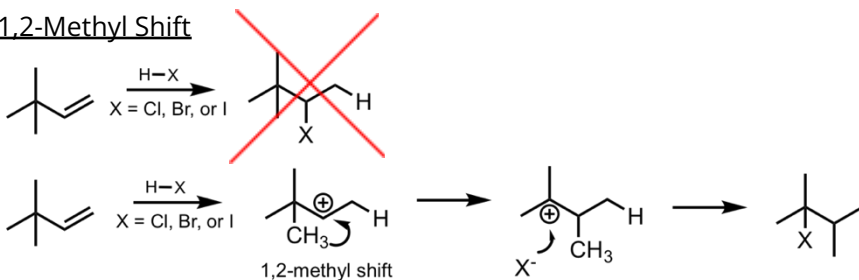
Rearrangements Details

When carbocations form, H's and CH₃'s can do a 1,2-shift to generate a more stable carbocation intermediate

1,2-Hydride Shift

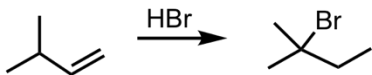


1,2-Methyl Shift



Alkene Reaction Details

Hydrohalogenation



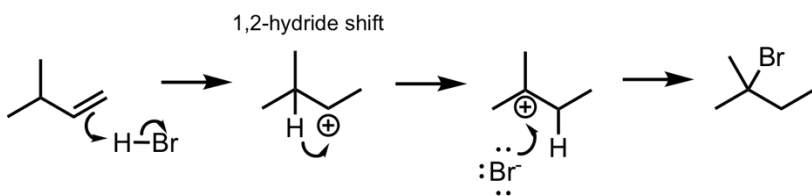
What's added: H⁺ and Br⁻

Regioselectivity: Markovnikov

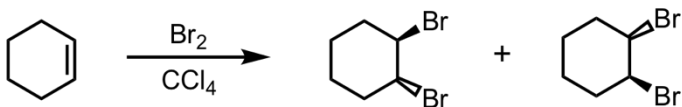
Intermediate: Carbocation

Rearrangement: Possible (methyl and hydride shifts)

Mechanism:



Halogenation



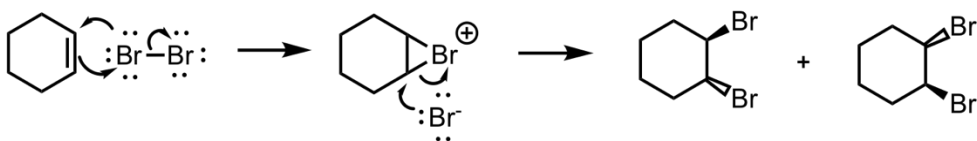
What's added: 2 Br atoms

Stereoselectivity: Anti

Intermediate: Bromonium ion

Rearrangement: Not possible

Mechanism:



Hydrobromination with Peroxide



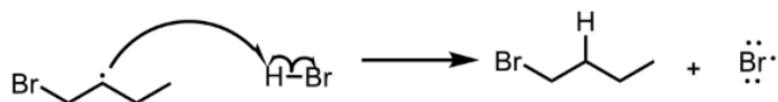
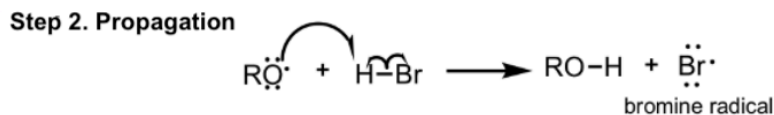
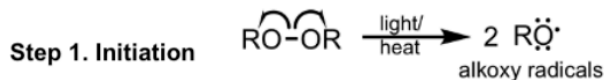
What's added: $\text{H}\cdot$ and $\text{Br}\cdot$

Regioselectivity: Anti-Markovnikov

Intermediate: Radical

Rearrangement: Not possible

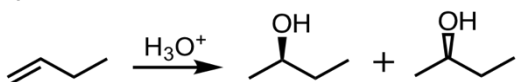
Mechanism:



Step 3. Termination



Hydration



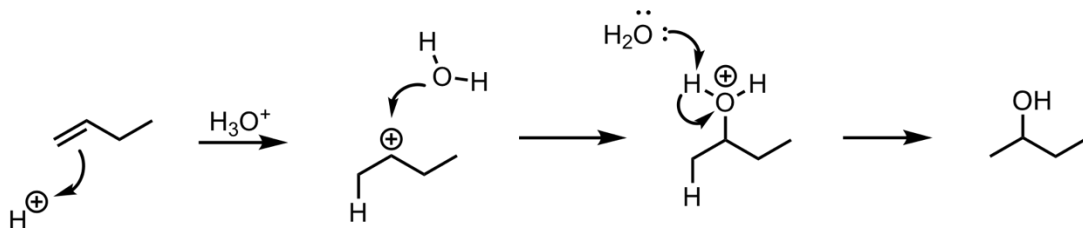
What's added: H^+ and OH^-

Regioselectivity: Markovnikov

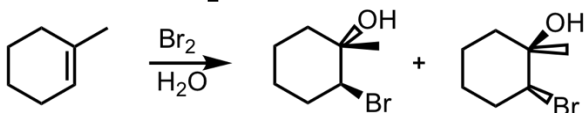
Intermediate: Carbocation

Rearrangement: Possible (methyl and hydride shifts)

Mechanism:



Bromination in H_2O



What's added: Br^+ and OH^-

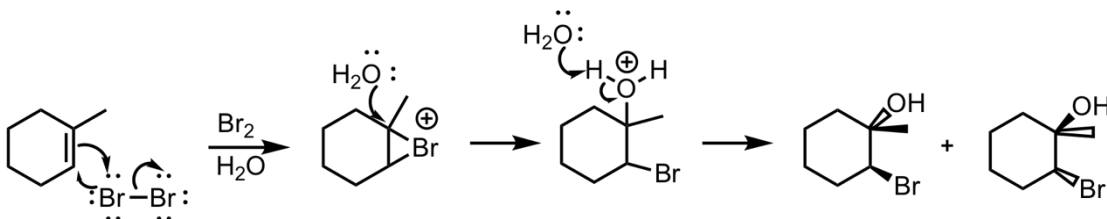
Regioselectivity: Markovnikov

Stereoselectivity: Anti

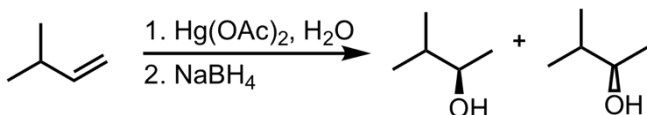
Intermediate: Bromonium ion

Rearrangement: Not possible

Mechanism:



Oxymercuration-Demercuration



What's added: H^+ and OH^-

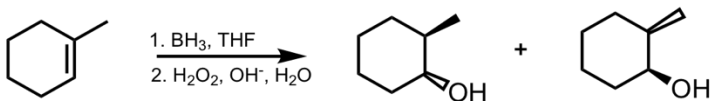
Regioselectivity: Markovnikov

Stereoselectivity: Anti

Intermediate: Mercurinium ion bridge

Rearrangement: Not possible

Hydroboration-Oxidation



What's added: H^+ and OH^-

Regioselectivity: Anti-Markovnikov

Stereoselectivity: Syn

Intermediate: Hydroxy-boranes

Rearrangement: Not possible

Syn-Dihydroxylation

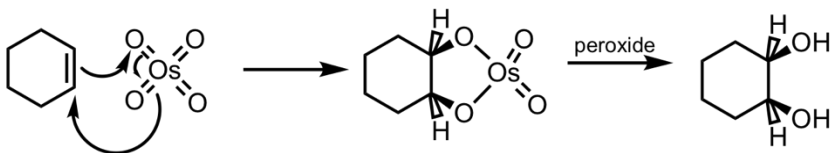


What's added: 2 OH groups

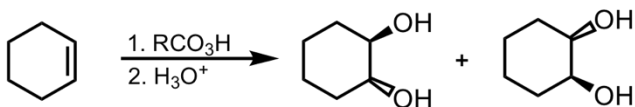
Stereoselectivity: Syn

Rearrangement: Not possible

Mechanism:



Anti-Dihydroxylation

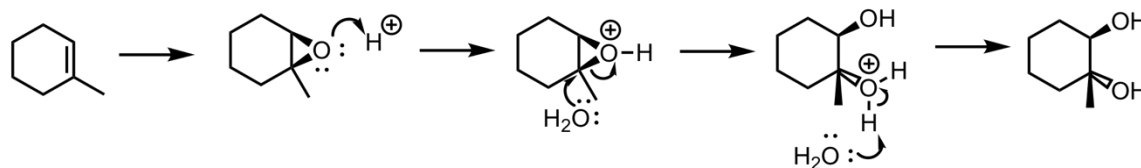


What's added: 2 OH groups

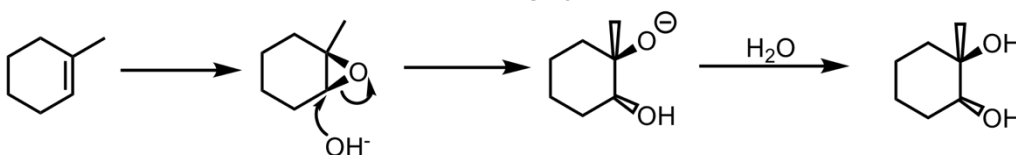
Stereoselectivity: Anti

Rearrangement: Not possible

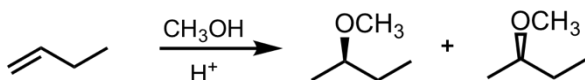
In acidic conditions, the H_2O attacks the more highly-substituted C:



In basic conditions, H_2O attacks the less highly-substituted C:



Addition of an Alcohol



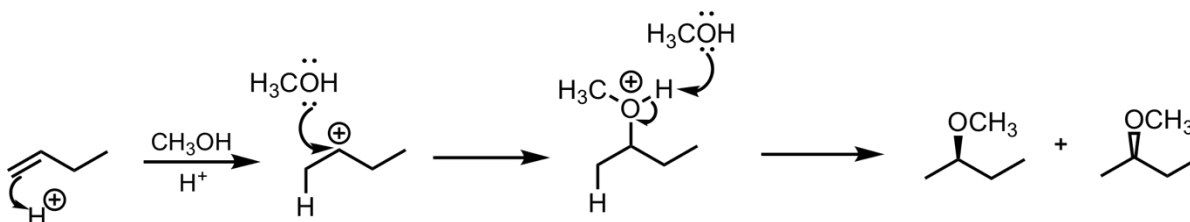
What's added: H^+ and OR^-

Regioselectivity: Markovnikov

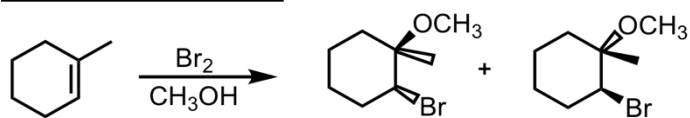
Intermediate: Carbocation

Rearrangement: Possible

Mechanism:



Bromination in Alcohol



What's added: Br^+ and OR^-

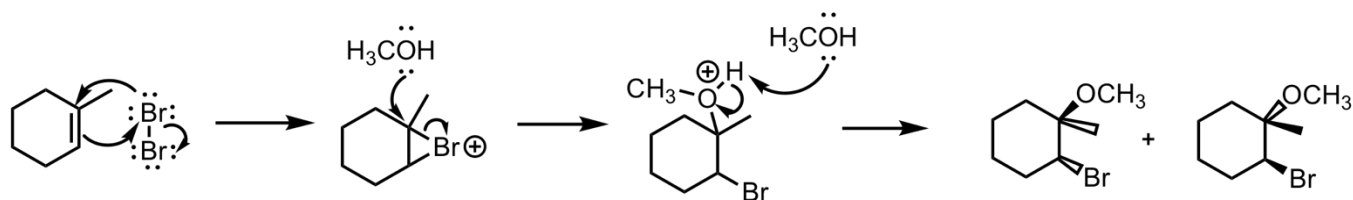
Regioselectivity: Markovnikov

Stereoselectivity: Anti

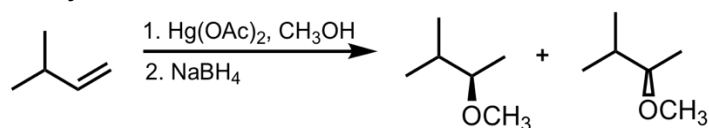
Intermediate: Bromonium ion

Rearrangement: Not possible

Mechanism:



Alkoxymercuration-Demercuration



What's added: H^+ and OCH_3^-

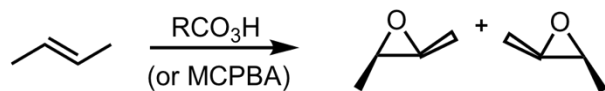
Regioselectivity: Markovnikov

Stereoselectivity: Anti

Intermediate: Mercurinium ion

Rearrangement: Not possible

Epoxidation

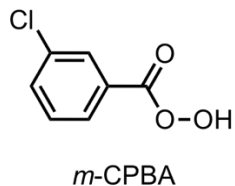


What's added: O

Stereoselectivity: Syn

Rearrangement: Not possible

Know that a commonly-used peroxy acid is *m*-CPBA:



Catalytic Hydrogenation



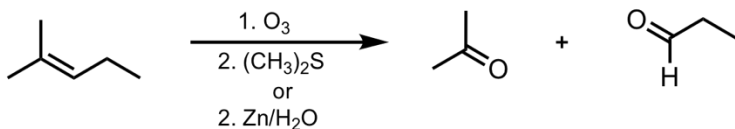
What's added: 2 H atoms

Stereoselectivity: Syn

Rearrangement: Not possible

Note: You may see Pt used as well. This is just the catalyst and does not change the outcome of the products.

Ozonolysis in Reducing Conditions



What's added: 2 O atoms

Regioselectivity: N/A

Stereoselectivity: N/A

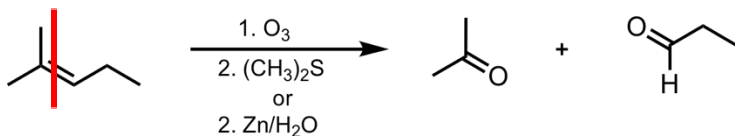
Intermediate: N/A

Rearrangement: N/A

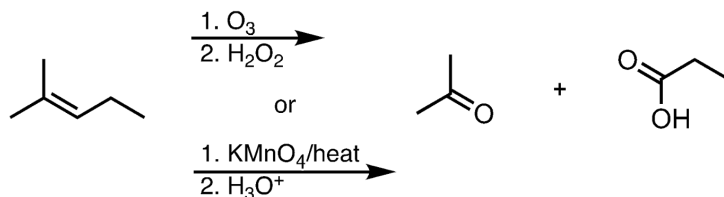
Mechanism: *You do not need to know the mechanism for this reaction*

Know that the $\text{C}=\text{C}$ double bond gets "sawed" in half, and an O atom is placed on the end of each new piece.

Note: $(\text{CH}_3)_2\text{S}$ is often abbreviated "DMS" for dimethyl sulfide.



Ozonolysis in Oxidizing Conditions/Oxidative Cleavage



What's added: Multiple O atoms

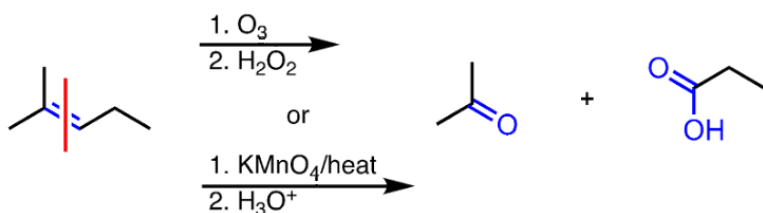
Regioselectivity: N/A

Stereoselectivity: N/A

Intermediate: N/A

Rearrangement: N/A

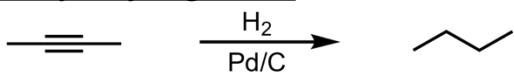
Mechanism: *You do not need to know the mechanism for this reaction*



Know that the $\text{C}=\text{C}$ double bond gets "sawed" in half, and an O atom is placed on the end of each new piece. Any H's attached to the alkene C's get replaced by an $-\text{OH}$ group since we are under oxidizing conditions/hot KMnO_4 . Unlike reducing conditions which would have formed aldehydes, oxidizing conditions produce carboxylic acids instead.

Alkyne Reaction Details

Catalytic Hydrogenation



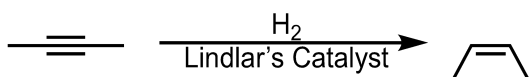
What's added: 4 H atoms

Stereoselectivity: Syn

Rearrangement: Not possible

Note: You may see Pt used as well. This is just the catalyst and does not change the outcome of the products.

Reduction to Cis-Alkene

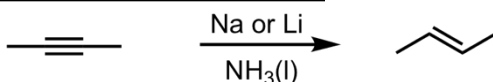


What's added: 2 H atoms

Stereoselectivity: Syn

Rearrangement: Not possible

Reduction to Trans-Alkene

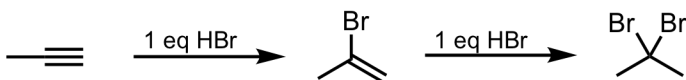


What's added: 2 H atoms

Stereoselectivity: Anti

Rearrangement: Not possible

Hydrohalogenation with HBr (Terminal Alkyne)



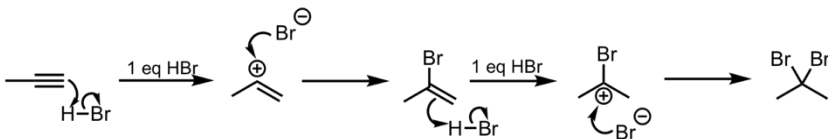
What's added: 1 H atom and 1 halogen atom (can be F, Br, I, or Cl) per equivalent of HX

Regioselectivity: Markovnikov

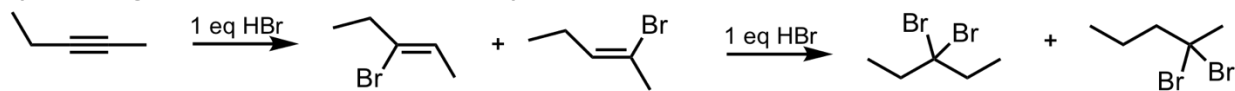
Intermediate: Carbocation

Rearrangement: Not possible

Mechanism: The halogen goes to the C with fewer H's



Hydrohalogenation with HBr (Internal Alkyne)



What's added: 1 H atom and 1 halogen atom (can be Cl or Br) per equivalent of HX

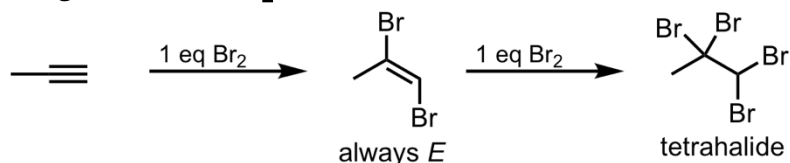
Regioselectivity: Markovnikov

Intermediate: Carbocation

Rearrangement: Not possible

Mechanism: Same as for terminal alkynes, but yields a mixture of two products because both intermediates are equally stable

Halogenation with Br₂



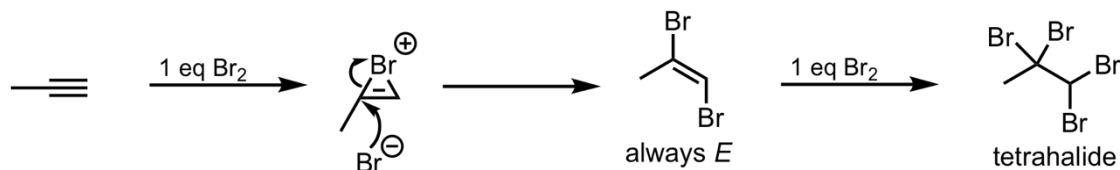
What's added: 2 halogen atoms (can be F, Br, I, or Cl) per equivalent of X₂

Stereoselectivity: Anti

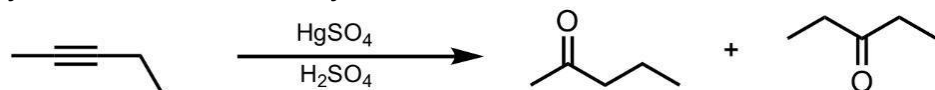
Intermediate: Bromonium ion

Rearrangement: Not possible

Mechanism:



Hydration of an Internal Alkyne



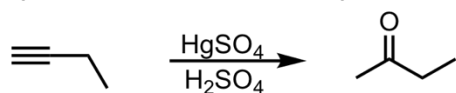
What's added: 1 O atom

Rearrangement: Not possible

Do know that this reaction produces enols, which then tautomerize to form ketones.



Hydration of a Terminal Alkyne (Markovnikov)

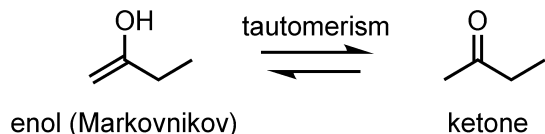


What's added: 1 O atom

Regioselectivity: Markovnikov

Rearrangement: Not possible

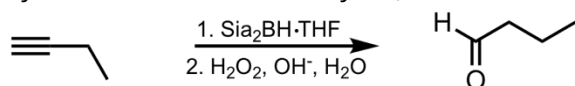
Know that this reaction produces Markovnikov enols, which then tautomerize to form ketones.



enol (Markovnikov)

ketone

Hydration of a Terminal Alkyne (Anti-Markovnikov)



What's added: 1 O atom

Regioselectivity: Anti-Markovnikov

Rearrangement: Not possible

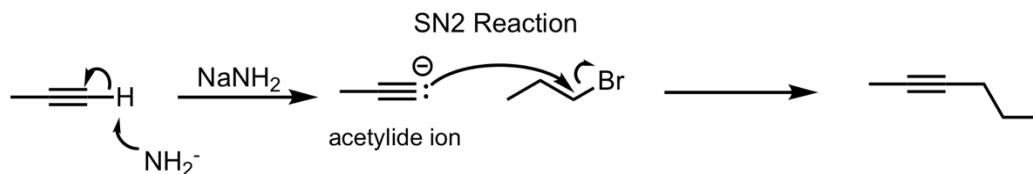
Know that this reaction produces Anti-Markovnikov enols, which then tautomerize to form aldehydes.



enol (Anti-Markovnikov)

aldehyde

SN2 Addition of an Acetylide Ion to an Alkyl Halide



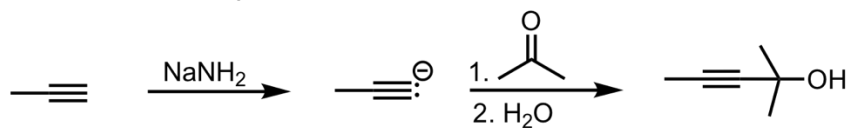
What's added: additional C atoms (-R of alkyl halide)

Intermediate: Acetylide Ion

Rearrangement: Not possible

Mechanism: Deprotonation, then alkylation via $\text{S}_\text{N}2$ reaction

Addition of an Acetylide Ion to a Ketone

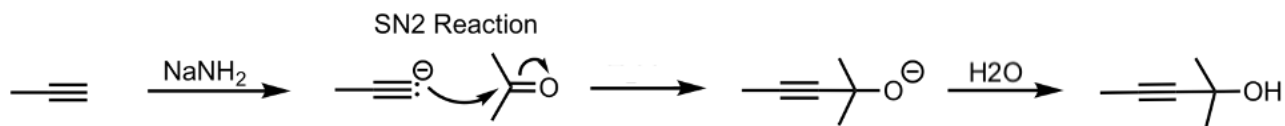


What's added: 1 alkyl group

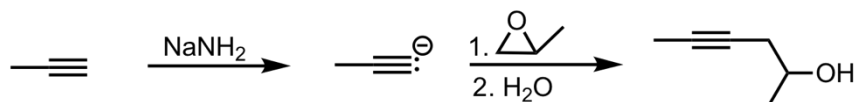
Intermediate: Acetylide Ion

Rearrangement: Not possible

Mechanism: Deprotonation, then addition of a ketone



SN2 Addition of an Acetylide Ion to an Epoxide

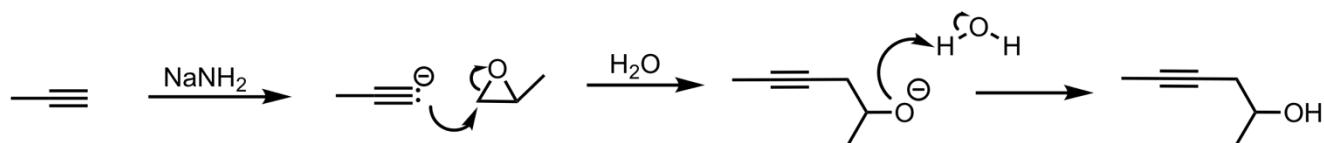


What's added: 2-hydroxylpropane (from epoxide)

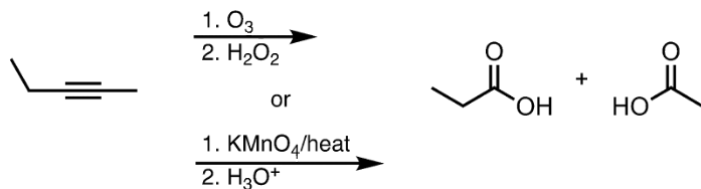
Intermediate: Acetylide Ion

Rearrangement: Not possible

Mechanism: Deprotonation, then addition of 2-hydroxyl propane via SN2 reaction

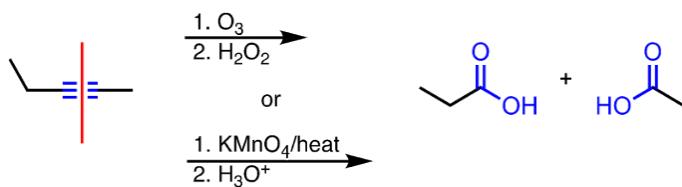


Ozonolysis/Oxidative Cleavage on an Internal Alkyne

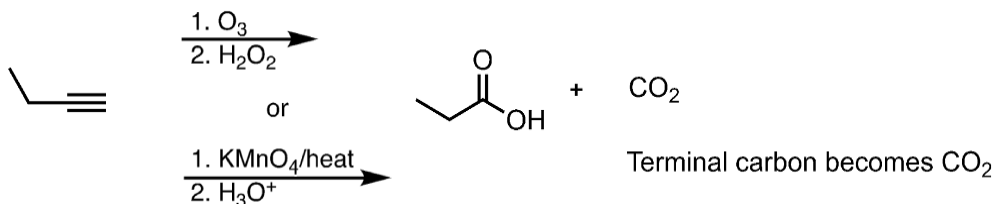


What's added: 4 O atoms and 2 H atoms

Know that the reaction cuts the triple bond in half. An O replaces two of the bonds as C=O and the third lone bond becomes a bond to -OH.

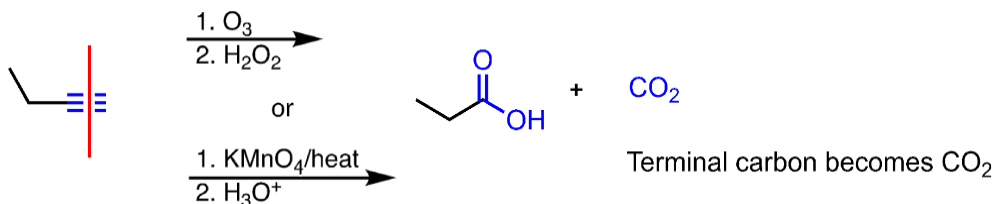


Ozonolysis/Oxidative Cleavage on a Terminal Alkyne



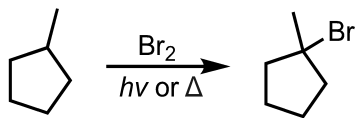
What's added: 4 O atoms and 1 H atom

Know that the reaction cuts the triple bond in half. On the internal side, an O replaces two of the bonds as C=O and the third lone bond becomes a bond to -OH. On the terminal side, two oxygens O replace all the bonds on carbon, forming the most oxidized form of carbon: CO_2 .



Free Radical Halogenation Reaction Details

Free Radical Halogenation using Bromine (more selective)



What's added: 1 Br atom

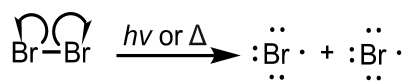
Regioselectivity: Most Substituted Product

Intermediate: Radical Intermediate

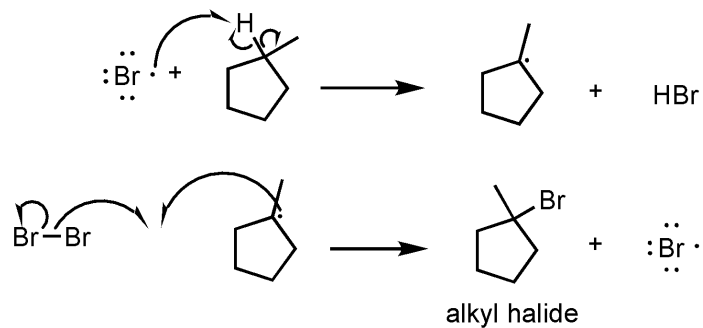
Rearrangement: Not possible

Mechanism: Formation of bromine and carbon radicals and them joining to create an alkyl halide

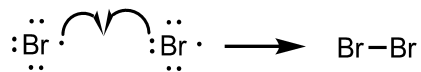
1. Initiation



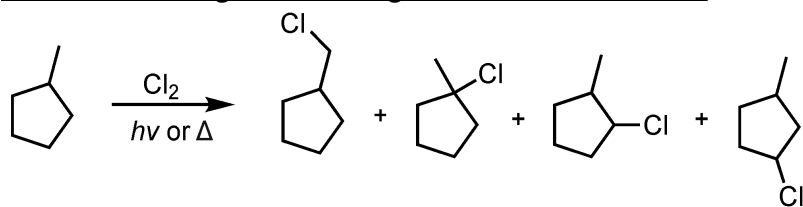
2. Propagation



3. Termination



Free Radical Halogenation using Chlorine (less selective)



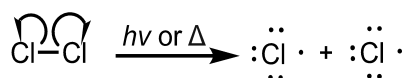
What's added: 1 Cl atom

Intermediate: Radical Intermediate

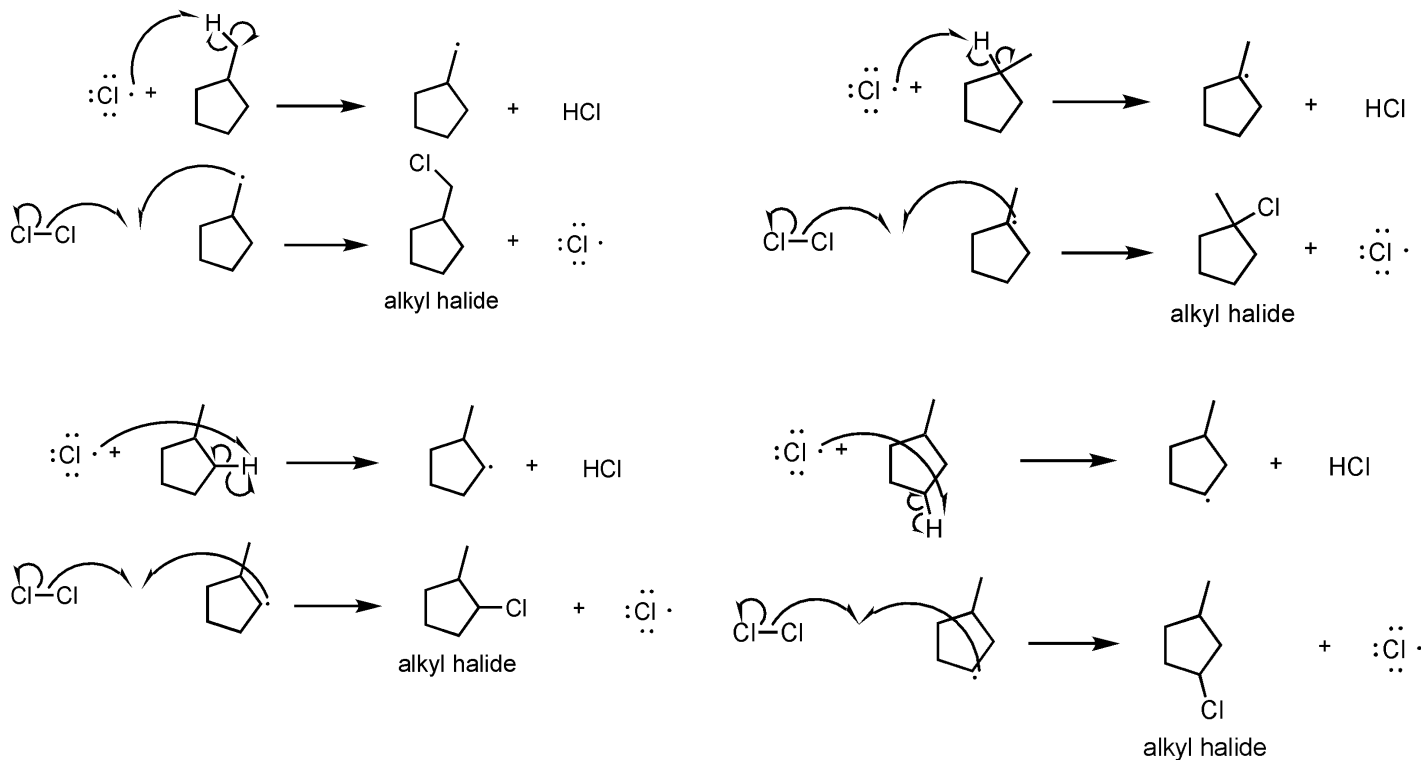
Rearrangement: Not possible

Mechanism: Formation of chlorine and carbon radicals and them joining to create alkyl halides

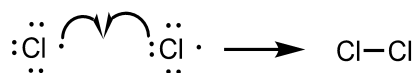
1. Initiation



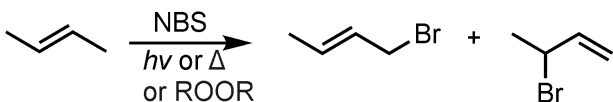
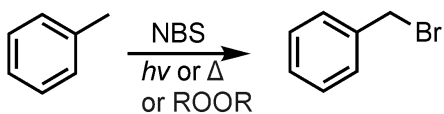
2. Propagation



3. Termination



Allylic/Benzylic Bromination

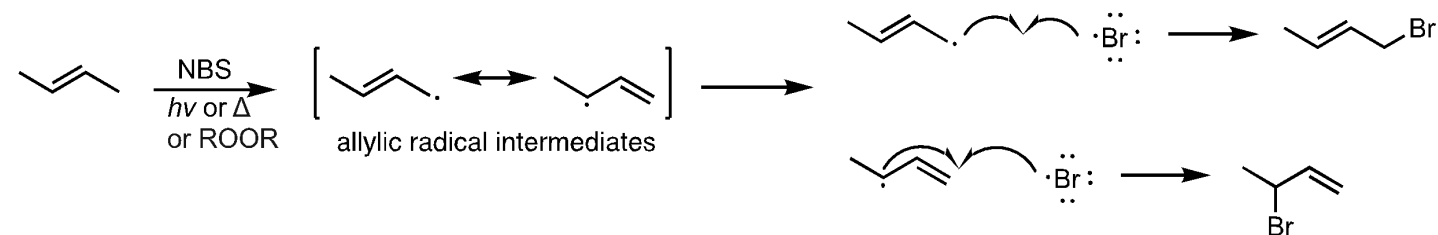


What's added: 1 Br atom

Intermediate: Allylic Radical Intermediate

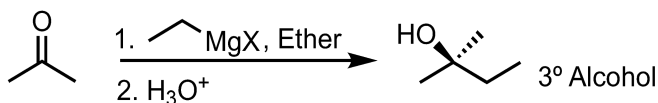
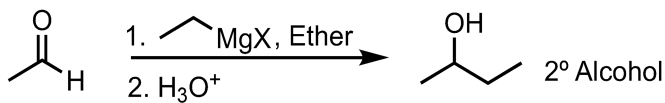
Rearrangement: Not possible

Note: this reaction results in the formation of allylic radical intermediates which resonate and thus allow for the formation of multiple products.



Grignard Reaction Details

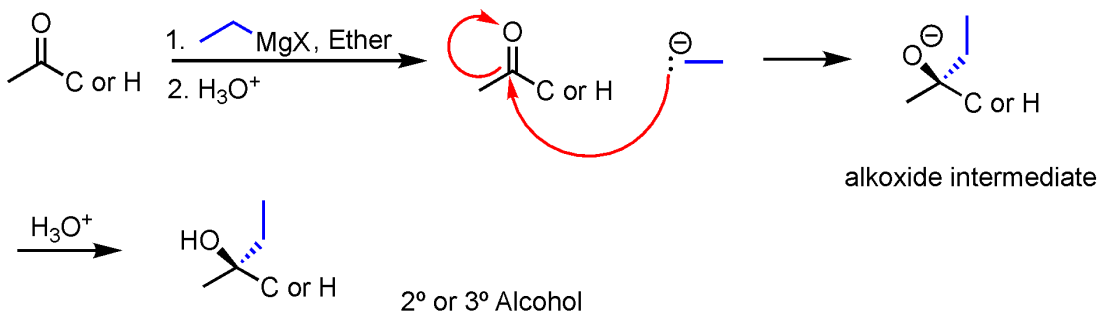
Addition of a Grignard Reagent to an Aldehyde/Ketone



What's added: R group

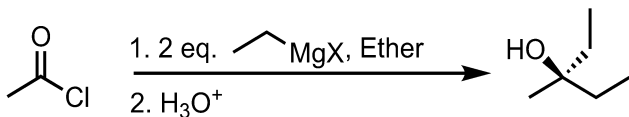
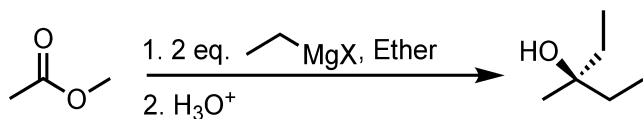
Intermediate: Alkoxide ion

Mechanism: Grignard reagents are electron-rich and attack electrophilic species such as carbonyls. The mechanism follows typical nucleophilic addition to a carbonyl, forming 2° or 3° Alcohols.



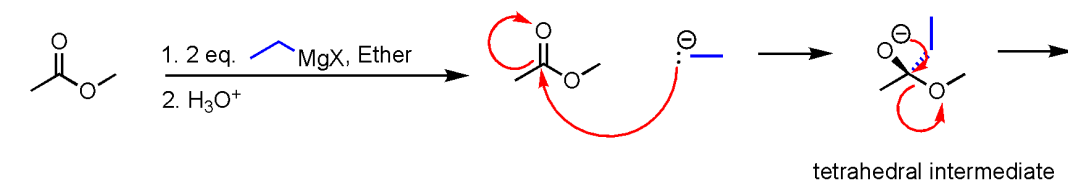
Note: Using R-MgX is virtually the same reaction as using R-Li (organolithium); you are using the R-group as a nucleophile to form an alcohol.

Addition of a Grignard Reagent to an Ester/Acyl Chloride

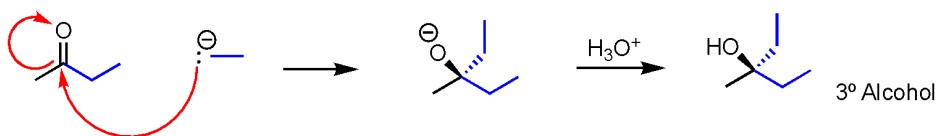


What's added: 2 R groups

Mechanism: Grignard reagents are electron-rich and attack electrophilic species such as carbonyls. The mechanism follows typical nucleophilic addition to a carbonyl with a twist; the Grignard reagent adds **twice** with esters and acyl chlorides, forming 3° alcohols. Here it is shown on an ester (the same mechanism as with an acyl chloride):

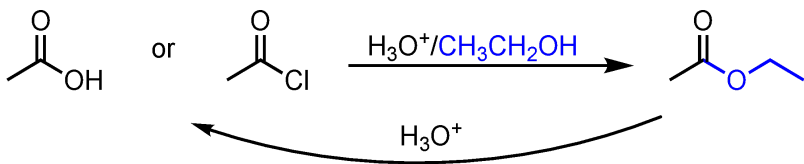


tetrahedral intermediate



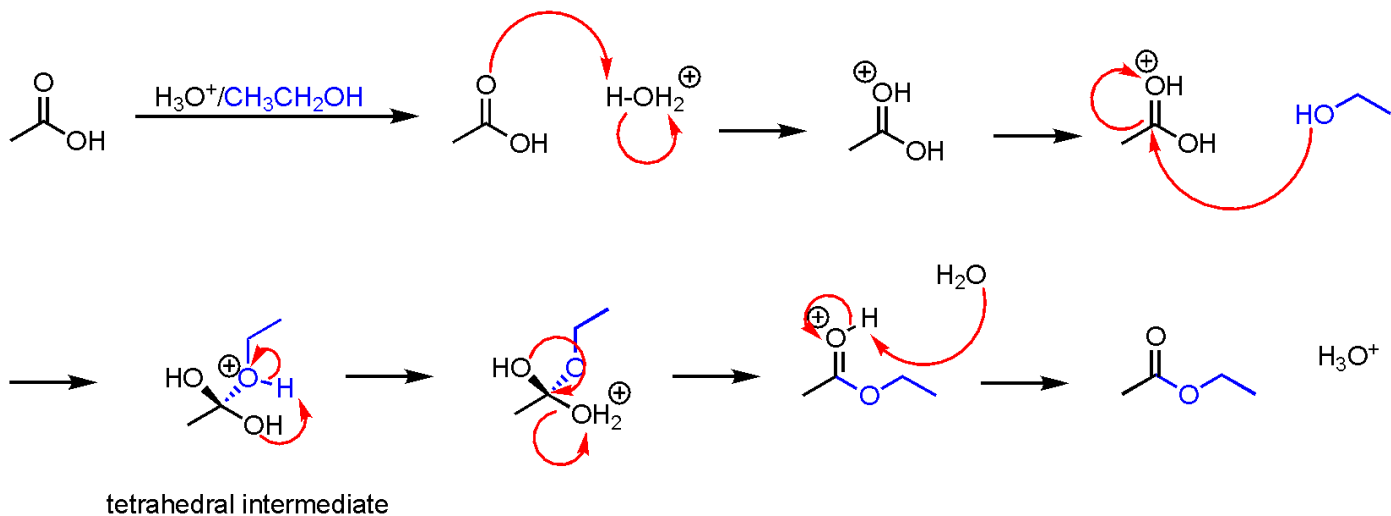
Carboxylic Acid Derivative Reaction Details

Fischer Esterification



What's added: An alcohol

Mechanism: Essentially, this reaction substitutes the Cl or OH of an acyl chloride/ester with an alcohol:



Notes: Notice that the reaction is reversible under H_3O^+ /heat. The reverse reaction is how esters are cleaved into their corresponding carboxylic acid and alcohol, an important biological process. Also, this is the same mechanism for any reaction involving a carboxylic acid derivative (anhydrides, amides, carboxylic acids, esters).